

stretchlim Function Test Results

Test Case 1: Default tolerance on uint8 ramp

Input (top-left 5x5):

0 1 2 3 4

Output:

Low = 0.0078431

High = 0.9921569

Test Case 2: Scalar tolerance [0.05]

Input (top-left 5x5):

0 1 2 3 4

Output:

Low = 0.0470588

High = 0.9529412

Test Case 3: Custom tolerance [0.20 0.80]

Input (top-left 5x5):

10 10 10 10 10

10 10 10 10 10

10 10 10 10 10

10 10 10 10 10

10 10 10 10 10

Output:

Low = 0.0392157

High = 0.9803922

Test Case 4: Constant uint8 image

Input (top-left 5x5):

100 100 100 100 100
100 100 100 100 100
100 100 100 100 100
100 100 100 100 100
100 100 100 100 100

Output:

Low = 0.0000000

High = 1.0000000

Test Case 5: Double precision image with varying intensities

Input (top-left 5x5):

0.1 0.1 0.1 0.1 0.1
0.1 0.1 0.1 0.1 0.1
0.1 0.1 0.1 0.1 0.1
0.1 0.1 0.1 0.1 0.1
0.1 0.1 0.1 0.1 0.1

Output:

Low = 0.1000000

High = 0.9500000

Test Case 6: Skewed histogram image

Input (top-left 5x5):

20 20 20 20 20
20 20 20 20 20
20 20 20 20 20
20 20 20 20 20
20 20 20 20 20

Output:

Low = 0.0784314

High = 0.9411765

Test Case 7: uint16 full-range image

Input (top-left 5x5):

0 256 512 768 1024

1 257 513 769 1025

2 258 514 770 1026

3 259 515 771 1027

4 260 516 772 1028

Output:

Low = 0.0099947

High = 0.9900053

Test Case 8: int16 image with varying intensity regions

Input (top-left 5x5):

-25000 -25000 -25000 -25000 -25000

-25000 -25000 -25000 -25000 -25000

-25000 -25000 -25000 -25000 -25000

-25000 -25000 -25000 -25000 -25000

-25000 -25000 -25000 -25000 -25000

Output:

Low = 0.1185321

High = 0.8814832

Test Case 9: RGB image with different channels

Input (top-left 5x5):

➔ **Red Channel:**

0 16 32 48 64

1 17 33 49 65

2 18 34 50 66

3 19 35 51 67

4 20 36 52 68

➔ **Green Channel:**

255 239 223 207 191

254 238 222 206 190

253 237 221 205 189

252 236 220 204 188

251 235 219 203 187

➔ **Blue Channel:**

128 128 128 128 128

128 128 128 128 128

128 128 128 128 128

128 128 128 128 128

128 128 128 128 128

Output:

Channel	Low	High
Red	0.0078431	0.9921569
Green	0.0078431	0.9921569
Blue	0.0000000	1.0000000

Test Case 10: Invalid tolerance

Input (first 10 elements):

0 1 2 3 4 5 6 7 8 9

Output: "stretchlim: TOL(1) must not exceed TOL(2)"

Additional Test Cases:

1) Input image:



Output:

Low = 0.0549020

High = 0.8627451

2) Input image:



Output:

Low = 0.3176471

High = 0.7921569

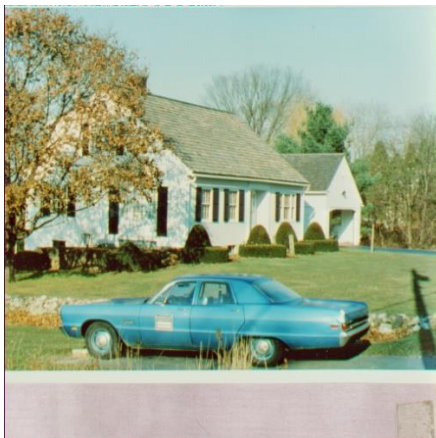
3) Input image:



Output:

Channel	Low	High
Red	0.1803922	0.8627451
Green	0.0862745	0.8823529
Blue	0.3098039	0.8627451

4) Input image:



Output:

Channel	Low	High
Red	0.145098	0.8784314
Green	0.0588235	0.8980392
Blue	0.054902	0.8509804